ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

ROBOTICS FUNDAMENTALS

PAPER 1 · PRACTICE C

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 2. Laid out like the DHET first paper. Section A (35): multiple choice, true or false, matching and choosing the correct word · Section B (65): electronic components, sensors and actuators, a Raspberry Pi doorbell program to study, Ohm's law and power calculations, and the laws of robotics. Formula sheet at the end.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B, and a formula sheet.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
6. Write neatly and legibly.

SECTION A

QUESTION 1

Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1–1.10) in the ANSWER BOOK.

1.1 Which 3D printing filament is the easiest for beginners because it prints at a low temperature and does not need a heated bed?

A PLA B ABS C Nylon D Polycarbonate

1.2 What does an ultrasonic sensor measure?

A Temperature B Light C Distance D Humidity

1.3 Which board is a single-board computer that runs a Linux operating system?

A Arduino Uno B Raspberry Pi
C ESP8266 module D 555 timer

1.4 What is the function of a heat sink?

A To store energy B To amplify signals
C To measure current D To carry heat away from a component

1.5 Which ONE of the following is an output device?

A Buzzer B Push button C PIR sensor D LDR

1.6 Which software turns a 3D model into layers and instructions for a 3D printer?

A Word processor B Slicer
C Web browser D Spreadsheet

1.7 In which unit is electrical power measured?

A Volt B Ohm C Watt D Ampere

1.8 A robot on Mars that is driven by an operator on Earth is an example of …

A full autonomy. B welding.
C pneumatics. D teleoperation (telepresence).

1.9 GPIO stands for …

A General Purpose Input/Output. B Ground Power Input/Output.
C Graphic Processing Input/Output. D General Program Interface Option.

1.10 Which law of robotics says that a robot must obey orders given by humans, except when this would harm a human?

A First law B Second law C Third law D Fourth law

(10)

[10]

QUESTION 2

Indicate whether the following statements are TRUE or FALSE. Write only 'True' or 'False' next to the question number (2.1–2.10) in the ANSWER BOOK.

2.1 The GPIO pins of a Raspberry Pi work at 3,3 V.

2.2 ABS filament gives off no fumes when it is printed.

2.3 Robots are well suited to repetitive tasks.

2.4 An LED is an input device.

2.5 A servo motor can be turned to a specific angle.

2.6 Voltage is measured in amperes.

2.7 A breakout board makes it easier to connect a small component to a breadboard.

2.8 Python code is case sensitive.

2.9 A hot 3D printer nozzle may be cleaned with bare fingers.

2.10 Ohm's law shows the relationship between voltage, current and resistance.

(10)

[10]

QUESTION 3

Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (3.1–3.10) in the ANSWER BOOK.

	COLUMN A		COLUMN B
3.1	Machine that builds an object layer by layer	A	resistance
3.2	Sensor that detects the movement of warm bodies	B	filament
3.3	Robot that works without a human controlling it	C	3D printer
3.4	Electric current that changes direction regularly	D	chassis
3.5	Opposition to the flow of current	E	alternating current (AC)
3.6	Frame on which the parts of a robot are mounted	F	gripper
3.7	Part of a robot arm that holds objects	G	PIR sensor

3.8	Plastic thread fed into a 3D printer	H	print bed
3.9	Pin on a board that can be programmed as an input or an output	I	autonomous robot
3.10	Flat surface on which a 3D print is built	J	GPIO pin

(10)

[10]

QUESTION 4

Complete the following sentences by choosing the correct word in brackets. Write only the word next to the question number (4.1–4.5) in the ANSWER BOOK.

4.1 A resistor is a (passive/active) component.

4.2 An ultrasonic sensor uses (sound/light) waves to measure distance.

4.3 A DC motor changes (electrical/chemical) energy into mechanical energy.

4.4 The (anode/cathode) is the longer leg of an LED.

4.5 A (servo/stepper) motor usually has a built-in potentiometer for feedback.

(5)

[5]

TOTAL SECTION A: [35]

SECTION B

QUESTION 5

5.1 Name the component described in each of the following:

5.1.1 A board full of holes, connected in rows, used to build circuits without soldering (1)

5.1.2 A two-legged component with coloured bands that limits current (1)

5.1.3 A small circuit board that carries a sensor chip and has pins so that it can plug into a breadboard (1)

5.1.4 A wire with a spring-loaded metal clip at each end (1)

5.1.5 A component that gives out light when current flows through it in the forward direction (1)

5.2 Explain why a breadboard is used when building a prototype. (3)

5.3 Explain the difference between a sensor and an actuator, and give ONE example of each. (4)

[12]

QUESTION 6

A student built a doorbell with a Raspberry Pi. A push button is connected to GPIO 17 and a buzzer to GPIO 27. When a visitor presses the button, the buzzer beeps and the program

counts the visitors.

Study the code below and answer the questions that follow.

```
 1  import RPi.GPIO as GPIO
 2  import time
 3
 4  BUTTON = 17
 5  BUZZER = 27
 6  GPIO.setmode(GPIO.BCM)
 7  GPIO.setup(BUTTON, GPIO.IN, pull_up_down=GPIO.PUD_UP)
 8  GPIO.setup(BUZZER, GPIO.OUT)
 9
10  presses = 0
11  try:
12      while True:
13          if GPIO.input(BUTTON) == GPIO.LOW:      # button pressed
14              presses += 1
15              print("Visitor number", presses)
16              for _ in range(3):
17                  GPIO.output(BUZZER, GPIO.HIGH)
18                  time.sleep(0.2)
19                  GPIO.output(BUZZER, GPIO.LOW)
20                  time.sleep(0.2)
21              time.sleep(1)                        # wait before the next press
22  except KeyboardInterrupt:
23      print("Doorbell stopped. Total visitors:", presses)
24  finally:
25      GPIO.cleanup()
```

6.1 Name the programming language and the library used to control the GPIO pins. (2)

6.2 What does pull_up_down=GPIO.PUD_UP in line 7 do, and why does the button read LOW when it is pressed? (3)

6.3 What does the statement presses += 1 in line 14 do? (3)

6.4 How many times does the buzzer beep for each press, and how long does one press keep the program busy (lines 16–21)? (3)

6.5 Why is there a time.sleep(1) in line 21? (3)

6.6 What happens when the user presses Ctrl + C while the program is running? (3)

6.7 What is the purpose of GPIO.cleanup() in line 25? (3)

6.8 Suggest TWO extra components that could improve this doorbell, and say what each would do. (3)

[23]

QUESTION 7

7.1 Define the term resistance. (2)

7.2 A 9 V battery is connected across a 450 Ω resistor. Calculate the current. (3)

7.3 A current of 0,5 A flows through a 24 Ω motor winding. Calculate the voltage across it. (3)

7.4 A robot charger supplies 5 V at 2 A. Calculate its output power. (3)

7.5 A 12 V lamp is rated at 36 W. Calculate its resistance. (3)

7.6 A 100 Ω resistor and a 220 Ω resistor are connected in series across a 9 V supply. Calculate the total resistance and the current. (4)

7.7 Explain the difference between a series circuit and a parallel circuit. (2)

[20]

QUESTION 8

Study the scenario below and answer the questions.

> A home-care robot looks after an elderly man and his grandchild. At night the man tells the robot to lock the front gate and switch off the outside lights. The robot's camera shows that the grandchild is still outside in the cold and cannot get back in once the gate is locked.

8.1 State the First Law of Robotics. (3)

8.2 Should the robot lock the gate immediately as it was told? Justify your answer with reference to the First and Second Laws of Robotics. (7)

[10]

TOTAL SECTION B: [65]

TOTAL: 100

END OF QUESTION PAPER

FORMULA SHEET

Speed: $v = d \div t$	Acceleration: $a = \Delta v \div \Delta t$
Force: $F = m \times a$	Weight: $W = m \times g$ ($g = 9{,}8\ m/s^2$)
Torque: $\tau = F \times r$	Work: $W = F \times s$
Ohm's law: $V = I \times R$	Power: $P = V \times I = I^2 \times R = V^2 \div R$
Energy: $E = P \times t$	Frequency: $f = 1 \div T$
Series: $R_T = R_1 + R_2 + R_3$; I is the same everywhere	Parallel: $1 \div R_T = 1 \div R_1 + 1 \div R_2$; V is the same across each branch